

Here is the written version of the image you provided:

"Statistics is the science which deals with the collection, classification, presentation, analysis and interpretation of numerical data."

Mode

Mode is the value that appears most frequently in a data set. It is one of the three main measures of central tendency, along with Mean and Median.

Mean

Mean is the average of a set of values. It is one of the most commonly used measures of central tendency, along with Median and Mode.

Median

Median is the middle value of a dataset when the values are arranged in ascending or descending order. It divides the data into two equal parts.

Types of Mean

1. **Arithmetic Mean (A.M.)** 
The most common type of mean — the sum of all values divided by the number of values.
 2. **Geometric Mean (G.M.)** 
The n-th root of the product of all values. Used for multiplicative relationships (like growth rates).
 3. **Harmonic Mean (H.M.)** 
Used when rates or ratios are involved (like speed, efficiency).
-

Standard Deviation

Standard Deviation measures how much the values in a dataset deviate from the mean.

Quartile Deviation

Quartile Deviation (also called Semi-Interquartile Range) measures the spread of the middle 50% of the data. It is half the difference between the third and first quartile.

Karl Pearson's Correlation

Karl Pearson's method measures the strength and direction of a linear relationship between two quantitative variables.

Frequency refers to the number of times a particular value or a group of values appears in a dataset.

A **class interval** is a group or range of values in which data is divided in a frequency distribution table.

Correlation is a **statistical measure** that shows the **strength and direction** of a relationship between two variables.

The **coefficient of correlation** is a **numerical value** (between -1 and +1) that shows how strongly two variables are related.

Regression analysis is a **statistical technique** used to **predict the value** of one variable (dependent variable) based on the value of another (independent variable).

Spearman's Rank Correlation

Spearman's rank correlation measures the strength of a monotonic relationship between two sets of ranks. It's used when data is ordinal or not normally distributed.



Scatter Diagram (Scatter Plot)

A **Scatter Diagram**, also known as a **Scatter Plot**, is a **graphical representation** used to show the **relationship between two quantitative variables**. Each point on the graph represents a **pair of values (X, Y)**.



Uses of Scatter Diagram:

- To **visualize the correlation** between two variables.
- To identify the **type of correlation** (positive, negative, or none).
- To detect **outliers** or unusual observations.

1. Karl Pearson's Correlation Coefficient

Also called **Pearson's r** , this method measures the **linear relationship** between two **continuous (quantitative)** variables.

Uses:

- When both variables are measured on **interval or ratio scales** (e.g., height, weight, income, marks).
- When the relationship between variables is expected to be **linear**.
- Common in scientific and business research where precision is required.

2. Spearman's Rank Correlation Coefficient

Also called **Spearman's ρ** , this method measures the **monotonic relationship** between two variables using **ranks** instead of raw data.

Uses:

- When data is **ordinal** (ranked) or **not normally distributed**.
- When the relationship is **non-linear but still monotonic** (i.e., consistently increasing or decreasing).
- Useful for **qualitative data** (like ratings, preferences, or ranks).